Statement of Work (SOW)

14 February 2025

- Dr. Mark Schoeberl (Chief Scientist, Science and Technology Corporation; mark.schoeberl@mac.com)
- Task Name: Aura Science Team and Atmospheric Composition Modeling and Analysis Program project on the Global Impact of Convection on the Upper Troposphere and Lower Stratosphere
- Period of Performance: 1 March 2025 – 28 February 2026
- Approval Name / NASA POC: Rei Ueyama (rei.ueyama@nasa.gov)

Task Descriptions

- Investigate the physical processes that control the abundance and distribution of cirrus clouds in the upper troposphere and lower stratosphere (UTLS) using a forward domain-filling Lagrangian model
    - Track the water vapor injected into the tropical upper troposphere by convection and link these convective events to the occurrence of cirrus clouds downstream
    - Quantify the fraction of observed stratiform cirrus clouds that could be directly attributable to convection by identifying the causal link between convective anvils and the downstream occurrence of stratiform clouds in the model
- Identify volcanic and pyrocumulonimbus aerosol plumes in SAGE III/ISS and OMPS-LP data via our unique cloud-aerosol separation technique and compare their UTLS signatures to those of deep convection
    - Use the Lagrangian model to link the dynamical dispersal of the aerosol plume with OMPS-LP observations and investigate the reason for the different behavior of the aerosol and water vapor plumes
    - Simulate the episodic injections of water vapor to quantify the long-term impact on UTLS clouds and water vapor budget
- Update and construct a more universal gravity wave spectrum based on recent airborne measurements including those over northern midlatitudes
    - Update the gravity wave parameterization in the model and examine the sensitivity of the results to these changes
- Submit results to peer-review journal